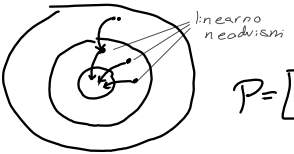
Jordanova formula

$\ker(A-\lambda I) \subset \dots \subset \ker(A-\lambda I)^m$

$\dim\ker(A-\lambda I)$ .... št celic za λ (geom. več)

m ..... velikost največje celice

$\dim\ker(A-\lambda I)^k - \dim\ker(A-\lambda I)^{k-1}$ .... št celic velikosti $k \times k$



$$P = \left[(A-\lambda I)^{m-1} v_1,\ (A-\lambda I)^{m-2} v_1, \dots, v_1,\ \dots\ v_2, \dots \right]$$

$$\mathcal{J} = \begin{bmatrix} J_1 & & \\ & \ddots & \\ & & J_n \end{bmatrix} \qquad f(\mathcal{J}) = \begin{bmatrix} f(J_1) & & \\ & \ddots & \\ & & f(J_n) \end{bmatrix}$$

$$f(J) = \begin{bmatrix} f(\lambda) & f'(\lambda) & \dots & \frac{f^{(n-1)}(\lambda)}{(n-1)!} \\ & \ddots & \ddots & \vdots \\ & & \ddots & f'(\lambda) \\ & & & f(\lambda) \end{bmatrix}$$

Vektorska polja

Tokovnica vektorskega polja je krivulja γ, ki reši $\dot{\gamma}(t) = F(\gamma(t))$

$F : D \to \mathbb{R}^n$

$\gamma : I \to \mathbb{R}^n$

Tok vektorskega polja je $\Phi : \mathbb{R}^n \times \underset{\sim +}{\mathbb{R}} \to \mathbb{R}^n$

$$\frac{\partial}{\partial t} \Phi_t(x) = F(\Phi_t(x)) \qquad \Phi_0(x) = x$$

Trditev: V vek. polje na $D \subseteq \mathbb{R}^n$. $V(p) = 0$ za nek $p \in D$

tokovnica γ ki zadošča $\gamma(0) = p$ $\quad \gamma \equiv p$

Po eksistenčnem izreku je to edina tokovnica

Linearni sistemi

Homogen sistem: $\dot{x} = Ax$

Fundamentalna rešitev: $\Phi(t) = e^{At} = Pe^{Jt}P^{-1}$

Splošna rešitev: $\Phi(t)x_0$

$\dot{\Phi} = A\Phi \qquad \Phi(0) = id$

Nehomogen sistem: $\dot{x} = Ax + f(t) \quad \left(\Phi(-t) = \Phi^{-1}(t)\right)$

Rešitev $x = x_n + x_p \qquad \dot{x}_n = Ax_n$

$x_n = e^{At}x_0 \qquad x_p(t) = \int_0^t \Phi(t)\Phi^{-1}(s) f(s)\,ds$

$x_p = e^{At}C(t) \rightsquigarrow e^{At}\dot{C}(t) = f(t)$

$C(t) = \int_0^t e^{-As} f(s)\,ds$

$x_p = e^{At}C(t) = e^{At}\int_0^t e^{-As} f(s)\,ds$

Nasveti:

- imaginarne lastne vrednosti? uvedba novih koordinat
- $x_0 = \sum v_i$; v_i lastni vektorji : nastavek: $x(t) = \sum e^{\lambda_i t} v_i$
- splošni nastavek $x_j(t) = \sum e^{\lambda_i t} p_i(t)$

  $p_i(t)$ polinom stopnje največ večkratnosti λ_i (v karakt. polinomu)

  (velja tudi za kompleksne

  (za rešitve linearnega sistema s konst. koeficienti)

Linearne enačbe n-tega reda

$$a_0 y^{(n)} + \dots + a_n y = 0$$

nastavek: $y(x) = e^{\lambda x}$

Dobimo $p(\lambda) = a_0 \lambda^n + \dots + a_n = 0$

$\lambda_1 \dots \lambda_n$ paroma različne $\Rightarrow$

$$y(x) = C_1 e^{\lambda_1 x} + \dots + C_n e^{\lambda_n x}$$

večkratna ničla (stopnje k) μ

$$b_0 e^{\mu x} + b_1 x e^{\mu x} + \dots b_{k-1} x^{k-1} e^{\mu x}$$

$\lambda = a + bi \rightsquigarrow Ae^{ax}\cos(bx) + Ce^{ax}\sin(bx)$

Eulerjeva LDE (reda n)

$$a_0 x^n y^{(n)} + \dots + a_n y = 0$$

nastavek: $x = e^t$

$z(t) = y(e^t)$

$$\Rightarrow a_0 z^{(n)} + \tilde{a}_1 z^{(n-1)} + \dots + \tilde{a}_n z = 0$$

$z(t) = p(t) e^{\lambda t} \rightsquigarrow y(x) = p(\ln x) x^{\lambda}$

$\lambda = a + bi \rightsquigarrow x^{\lambda} = Ax^a \cos(\ln b) + Bx^a \sin(b \ln x)$

Nehomogene enačbe

$$a_0 y^{(n)} + \dots + a_n y = e^{\mu x} p \qquad \text{rešitev: } y_h + y_p$$

y_p rešitev $\Rightarrow$ vse rešitve so oblike $y_h + y_p$

nastavek $y_p = e^{\mu x} g(x) x^k$

g ... polinom stopnje kot p

k .... kratnost μ v $p(\lambda)$ (al: 0 če μ ni ničla)

Eulerjeva DE

$$a_0 x^n y^{(n)} + \dots + a_n y = x^{\mu} p(\ln x)$$

Nastavek: $y_p = x^{\mu} g(\ln x)(\ln x)^k$

g ... iste stopnje kot p

k .... kratnost μ v $P(\lambda)$

Variacija konstante za LDE

$$y_h = C_1 y_1(x) + \dots + C_n y_n$$

$$y_p = C_1(x) y_1(x) + \dots + C_n(x) y_n(x)$$

odvodi od c_j-jev

$$\Phi \begin{bmatrix} c_1' \\ \vdots \\ c_n' \end{bmatrix} = \begin{bmatrix} y_1 & \dots & y_n \\ y_1' & & y_n' \\ \vdots & & \vdots \\ y_1^{(n-1)} & \dots & y_n^{(n-1)} \end{bmatrix} \begin{bmatrix} c_1' \\ \vdots \\ c_n' \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ f/a_0 \end{bmatrix}$$

Determinanta Wronskega

$$W(t) = \det \Phi(t) = \begin{vmatrix} y_1 & y_2 & \dots & y_n \\ y_1' & \dots & \dots & y_n' \\ \vdots & & & \\ y_1^{(n-1)} & \dots & \dots & y_n^{(n-1)} \end{vmatrix}$$

Liouvillov izrek: $W'(t) = tr(A(t))W(t)$

$$W(t) = W(t_0)\, e^{-\int_{t_0}^{t} tr(A(s))\,ds}$$

u, v lin neod. rešitvi $y'' + p(x)y + g(x)y = 0 \Rightarrow W = -u'v - uv'$

$$= \begin{vmatrix} u & v \\ u' & v' \end{vmatrix}$$

$W' + pW = 0 \rightarrow W = e^{-\int p\,dt}$

Če sta u, v lin neodvisni, v eni točki sta linearno neodvisni.

Vemo: $y'' + py' + gy = 0$ p, g zvezni na $I \subset \mathbb{R}$ intervalu

1. $\Rightarrow$ vse ničle od rešitve u so enostavne
2. $\Rightarrow$ množica ničel u nima stekališča v I
3. $\Rightarrow x_1 < x_2$ $u(x_1) = u(x_2) = 0 \Rightarrow \exists x_3 \in (x_1, x_2).\ v(x_3) = 0$

Vemo: $y'' + py' + gy = 0$

1) $W' + pW = 0$
2) $W(x) = W(x_0)\, e^{-\int_{x_0}^{x} p(t)\,dt}$ p, g zvezni funkciji
3) $v = u \int \frac{W}{u^2}\,dx$

VARIACIJSKI RAČUN

$A \leq \mathcal{C}^2([a,b])$

$\mathcal{L} : A \longrightarrow \mathbb{R}$ $\qquad L : [a,b] \times \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}$ $\quad C^2$

$$\mathcal{L}(y) := \int_a^b L(x, y(x), y'(x))\,dx$$

Smerni odvod: $D\mathcal{L}_y(h) = \frac{d}{dt}\Big|_{t=0} \mathcal{L}(y+th)$

$$Q\mathcal{L}_y(h) = \frac{d^2}{dt^2}\Big|_{t=0} \mathcal{L}(y+th)$$

$F : \mathbb{R}^n \longrightarrow \mathbb{R} \in C^2 \Longrightarrow DF = \nabla_h F(x)$

$$QF = h^T H_F(x) h = \langle h, H_F(x) h \rangle$$

$\mathcal{A} := \{ y \in C^2([a,b]) ; y(a) = A, y(b) \in B \}$

$y + th \in \mathcal{A} \Longrightarrow h(a) = 0 \ \wedge \ h(b) = 0$

$$D\mathcal{L}_y(h) = \int_a^b \left(L_y(x,y,y') - \frac{d}{dx} L_{y'}(x,y,y') \right) h(x)\,dx$$

y_0 je ekstrem $\Longleftarrow D\mathcal{L}_y(h) = 0 \Longleftrightarrow L_y - \frac{d}{dx} L_{y'} \equiv 0$

Euler-Lagrangeev pogoj

$L = L(y, y')$ $\qquad \mathcal{L}(y)$ ima kritično točko v $y \Longleftrightarrow L - y' L_{y'} \equiv C \in \mathbb{R}$

Bertramijeva identiteta

$$Q\mathcal{L}_{y_0}(h) = \int_a^b \left(h^2 \left(L_{yy} - \frac{d}{dx} L_{yy'} \right) + h'^2 L_{y'y'} \right) dx$$

y_0 kritična $\wedge$ $Q\mathcal{L}_{y_0}(h) > 0 \Longrightarrow y_0$ je minimum

y_0 kritična $\wedge$ $Q\mathcal{L}_{y_0}(h) < 0 \Longrightarrow y_0$ je maksimum

Keristno

$$C = \int \frac{1}{(1+x^2)^2} dx \rightsquigarrow C = A\arctan x + B\ln(1+x^2) + \frac{Px+E}{1+x^2}$$

$$ch^2 t = \frac{ch 2t + 1}{2}$$

$$ch^2 t - sh^2 t = 1$$

$$ch 2t = ch^2 t + sh^2 t$$

$$sh 2t = 2 sh t ch t$$

$$ar sh t = \ln|x + \sqrt{1+x^2}|$$

Nehomogeni sistemi s konst. koeficienti

$$\dot{x} = Ax + f(t)$$

homogeno: $\dot{x} = Ax$

$$\Phi(t) = e^{At} = Pe^{Jt}P^{-1}$$

$$x_h = \Phi(t)x_0$$

nehomogeno:

$$x_p(t) = \int_0^t \Phi(t)\Phi^{-1}(s)\cdot f(s)\,ds =$$

$$= Pe^{Jt}\int_0^t e^{-Js} f(s)\,ds$$

Z nekonstantnimi koeficienti

$\dot{x} = A(t)x$

imamo eno rešitev $\begin{bmatrix} x_h \\ y_h \end{bmatrix}$ iščemo drugo rešitev

s komponentama u in v $\begin{bmatrix} u \\ v \end{bmatrix}$

$$\Phi = \begin{bmatrix} x_h & u \\ y_h & v \end{bmatrix} \qquad W = \det \Phi = x_h v - u y_h$$

$$W = W(t_0)\, e^{\int_{t_0}^{t} \operatorname{tr}(A(s))\, ds}$$

$\dot{x} = A(t) + f(t)$ $\rightsquigarrow$ variacija konstante

$$\Phi \begin{bmatrix} A' \\ B' \end{bmatrix} = f(t)$$

$$x_p = A \begin{bmatrix} x_h \\ y_h \end{bmatrix} + B \begin{bmatrix} u \\ v \end{bmatrix}$$